

GapFade Algo

Operations Runbook & Daily Schedule

Aarav Pant — May 2026 — v1.1

System Overview — Two parallel strategies run fully automated via Windows Task Scheduler. The **Swing algo** scans 65+ symbols on daily bars for multi-day momentum trades (hold 3–15 days). The **Intraday algo** scans the same universe on 5-min bars for same-day CALL/PUT signals (hold minutes to hours). Both log paper trades to JSON files and validate P&L nightly. Push notifications fire on Grade S/A signals via ntfy.sh.

Daily Automated Schedule

Time (ET)	Task Name	Script	What It Does
Login	GapFadeAlgo	api.py	Starts the web dashboard + intra-day scanner. Loads config.env (ntfy topic).
9:30 AM	GapFadeAlgo	scripts/daily_scan.py	Swing scan: scores all symbols on daily bars, logs CALL/PUT signals to paper_trades.json.
Continuous	api.py scanner	day_trading.py	Every 5 min: pre-screens top 45 symbols by rel-vol, scores on 5-min bars, fires ntfy push for Grade S/A.
3:45 PM	GapFadeAlgo	scripts/daily_scan.py --close	Swing close scan: checks open swing trades, marks exits.
4:00 PM	GapFadeAlgo	scripts/daily_validate.py	Fetches real P&L for closed swing trades, prints win rate / profit factor.
4:15 PM	GapFadeAlgo	scripts/intraday_validate.py --close-eod	Closes open intraday trades via EOD daily OHLC simulation, logs accuracy stats.
Friday 4:30 PM	GapFadeAlgo	scripts/weekly_review.sh	Runs all three validators + threshold optimizer, prints full week summary.

Days that run: Swing tasks run Mon–Fri. Intraday validate runs Mon–Fri. Weekly review runs **Friday only**. The scanner (api.py) runs whenever you start it – it self-guards against after-hours signals using a wall-clock ET check.

Weekly Manual Checklist (5 min every Friday)

1. **Read the weekly review output** — check validate.log or re-run: `bash scripts/weekly_review.sh`

2. **Check the threshold recommendation** from `scripts/optimize_threshold.py`. If it suggests a new threshold, update `day_trading.py`: `SIGNAL_THRESHOLD = XX`
3. **Scan for any Grade S/A signals** you may have missed. Open the web dashboard (`python3 api.py` then `http://localhost:5000`).
4. **Review swing trades** still open: check `data/paper_trades.json` for trades open longer than 15 days — consider manually closing.
5. **Review intraday accuracy**: `data/accuracy_stats.json` — track win rate trend week over week.

Key Files Reference

File	Purpose
<code>data/day_trades.json</code>	All intraday paper trades (open + closed)
<code>data/paper_trades.json</code>	All swing paper trades (open + closed)
<code>data/accuracy_stats.json</code>	Historical accuracy runs (win rate, avg P&L, profit factor)
<code>data/config.env</code>	Push notification topic (<code>NTFY.TOPIC=aarav-scanner-9274</code>)
<code>data/scanner.log</code>	Rolling log from <code>api.py</code> scanner (5 MB × 3 backups)
<code>data/validate.log</code>	Output from each scheduled validate run
<code>data/portfolio.json</code>	Paper cash balance (used for position sizing)
<code>scripts/intraday_validate.py</code>	Close-EOD simulator & accuracy reporter
<code>scripts/optimize_threshold.py</code>	Finds best signal threshold from closed trade history
<code>scripts/intraday_backtest.py</code>	Walk-forward backtest on 2 years of hourly bars
<code>scripts/weekly_review.sh</code>	Friday master report script
<code>day_trading.py</code>	Intraday scanner: indicators, scoring engine, signal grader
<code>core_signals.py</code>	Shared math: RSI, EMA, CMF, OBV, MFI, BB
<code>Backtest.py</code>	Walk-forward backtest for the swing algo

Signal Grading System

Grade	Score	ntfy Push	Meaning
S	≥ 90	Yes (High priority)	Exceptional confluence — strongest possible setup
A	≥ 80	Yes (Normal)	High conviction — all major factors aligned
B	≥ 68	No	Above threshold — take selectively
C	≥ 55	No	Below threshold — logged but not actioned
D	< 55	No	Noise — ignored

Current threshold: 68 (Grade B and above fire signals).

Trade management: Stop = entry $\pm$ ATR $\times$ 2.0 Target = entry $\pm$ ATR $\times$ 3.0 Minimum R:R = 1.5

Scoring Engine — What Gives Points

Signal Factor	Points
EMA9 > EMA21 <i>and</i> price above VWAP (full trend)	+20
Price above/below VWAP	+8
15-min HTF EMA uptrend / downtrend confirmation	+10
RSI momentum (> 58 bull / < 42 bear)	+15
ADX > 30 (strong trend)	+15
15-bar high breakout / low breakdown	+15
Opening Range Breakout (ORB)	+25
Momentum continuation (price + EMA9 + volume)	+10
EMA9 cross-up / cross-down reversal	+20
Liquidity sweep + engulfing candle	+15
Explosive candle ($> \text{ATR} \times 0.4$)	+15
Volume > 2 $\times$ average	+18
Full smart money confluence (vol + CMF + OBV + MFI)	+18
Trend day setup (ADX > 30 + VWAP + EMA aligned)	+20
RS vs SPY > 2% outperformance	+12
RSI exhausted (> 78 overbought)	-10 to -20
Midday chop window (11am-1pm ET)	-15
Too extended from EMA9 ($> \text{ATR} \times 1.5$)	-20
Liquidity sweep trap (bull/bear trap)	-10
Bear market regime (SPY trend down)	-15

Universe & Pre-Screener

The scanner starts with **65 symbols** across 9 sectors. Before full evaluation, a parallel pre-screener fetches relative volume for all symbols in ~ 3 seconds and keeps only the **top 45 by relative volume** (minimum rel-vol 0.6 $\times$). This reduces noise and focuses compute on the most active names each session.

Sectors covered: Index ETFs (SPY, QQQ, IWM, TQQQ) — Mega-cap tech (TSLA, NVDA, AAPL, MSFT, META, GOOGL, AMZN) — Semiconductors (NVDA, AMD, INTC, QCOM, MU) — Financials (JPM, GS, COIN) — Energy (XOM, CVX, OXY) — Healthcare (LLY, MRNA) — Consumer/Internet (UBER, ABNB, ROKU) — High-beta / Speculative (MARA, RIOT, PLTR, SOFI, RIVN, UPST).

Performance Benchmarks — What to Look For

Metric	Fail	Borderline	Pass	Notes
Win Rate	< 45%	45–55%	> 55%	Aim > 55% on Grade A/S
Profit Factor	< 1.0	1.0–1.3	> 1.3	Gross wins ÷ gross losses
Avg P&L/trade	< -0.1%	-0.1 to 0%	> 0%	After 0.10% slippage
Expectancy $\times \sqrt{n}$	< 0	0–2	> 2	Used by optimizer

Current live results (as of May 2026): Win rate 50%, Profit Factor 1.78 on closed intraday trades. This is early data — aim for **3–4 weeks** (50+ closed trades) before drawing conclusions. Grade S/A signals are the primary focus; Grade B is secondary.

Useful Commands

Command	Does
<code>python3 api.py</code>	Start scanner + web dashboard (localhost:5000)
<code>python3 day_trading.py --once</code>	Single snapshot scan, prints to terminal
<code>python3 scripts/intraday_validate.py --close-eod</code>	Close open intraday trades + print accuracy
<code>python3 scripts/intraday_validate.py</code>	Just print accuracy (no closing)
<code>python3 scripts/optimize_threshold.py</code>	Find best signal threshold
<code>python3 scripts/intraday_backtest.py</code>	2-year walk-forward backtest (hourly proxy)
<code>python3 Backtest.py</code>	Walk-forward backtest for swing algo
<code>bash scripts/weekly_review.sh</code>	Full Friday report
<code>python3 day_trading.py --symbols TSLA NVDA</code>	Scan specific symbols only

Push Notifications Setup

1. App: **ntfy** (free, no account) — iOS App Store / Google Play
2. Your topic: `aarav-scanner-9274` (configured in `data/config.env`)
3. Fires on **Grade S** (high priority) and **Grade A** (normal) signals only
4. Format: `TSLA CALL Grade S score=94 entry=$312.40 stop=$308.20 target=$318.90`

5. To test: run `python3 day_trading.py --once` during market hours

Next 3–4 Weeks: What to Track

Week 1: Validate data pipeline. Confirm `validate.log` is updating at 4:15 PM each day. Check that closed trade P&Ls make sense (not all 0%). Monitor ntfy alerts.

Week 2: First performance read. After 20+ closed intraday trades, check win rate and profit factor. Run `optimize.threshold.py` — adjust threshold if recommended.

Week 3: Grade breakdown. Separate Grade S/A from Grade B trades. If S/A $WR \geq 60\%$, the high-conviction filter is working.

Week 4: Decision point. With 50+ closed trades: does the algo have a confirmed edge? If $WR > 55\%$ and $PF > 1.3$ — consider whether to adjust position sizing or universe.

Remember: The intraday backtest on hourly bars showed no edge because hourly bars $\neq$ 5-min intraday bars. The live paper trade data over the next 3–4 weeks is the real signal. Do not tune the algo based on fewer than 30 closed trades — that is not enough data to be statistically meaningful.